American International University-Bangladesh (AIUB)
Department of Computer Science
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Section: B

Software Quality Assurance and Testing

AGRI-X

A Report submitted
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Software Test Plan

for

AGRI-X

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American International University-Bangladesh

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1. TEST PLAN IDENTIFIER: AGRIX-TP-V1.0

Project Name: AGRI-X – AI-Based Smart Agriculture Platform

2. REFERENCES

I. Software Requirement Specification (SRS) – AGRI-X v2.0
II. SQAT Course Material – AIUB
III. IEEE 829 Standard for Software Test Documentation

3. PROBLEM DOMAIN

3.1 Background to the Problem

Bangladesh is an agricultural country, with nearly half of its workforce directly or indirectly dependent on farming for their livelihoods. Farmers across the country grow a wide variety of crops including rice, jute, vegetables, fruits, and spices. These crops not only feed the nation but also contribute significantly to the economy. However, despite the importance of agriculture, farmers face severe challenges that limit their productivity, income, and overall well-being.

One of the most critical problems is the lack of early detection of crop diseases. Every year, diseases like rice blast, leaf blight, and fruit rot destroy thousands of hectares of crops before farmers can properly identify them. Most farmers rely on traditional knowledge, neighboring advice, or trial-and-error methods rather than scientific diagnosis. By the time a disease is visibly widespread, it is often too late to prevent major yield loss.

Another major issue is the presence of middlemen in the agricultural supply chain. Farmers sell their products to local intermediaries at very low prices. These middlemen then sell the same products to city markets or exporters at much higher prices, keeping most of the profit for themselves. Farmers remain poor despite working hard, while buyers in cities pay high prices without knowing how little the farmer actually received.

Additionally, farmers have limited access to reliable weather forecasts, soil health data, and expert agricultural advice. Climate change has made farming more unpredictable, with unexpected rainfall, droughts, and temperature changes damaging crops. Without real-time environmental data, farmers cannot make informed decisions about irrigation, fertilizer use, or harvesting timing.

The root cause of these problems is the absence of an integrated, affordable, and easy-to-use digital platform that combines **AI-driven crop disease detection**, **real-time environmental data**, and a **direct farmer-to-buyer marketplace**. While some agricultural apps exist in Bangladesh, they are either too expensive, too complex, or focus on only one aspect such as weather or marketplace, leaving farmers without a complete solution.

This problem is extremely important to address because agriculture is the backbone of Bangladesh's economy and food security. Millions of small and marginal farmers depend on farming for their survival. Crop losses due to diseases and unfair pricing push farming families deeper into poverty. Moreover, sustainable and eco-friendly farming practices cannot be promoted without proper technological support. Without a smart agricultural platform like AGRI-X, farmers will continue to suffer from preventable crop losses, exploitation by middlemen, and lack of access to modern farming knowledge.

3.2 Solution to the Problem

The **AGRI-X** is an AI-powered smart agriculture platform designed specifically for Bangladeshi farmers and agricultural buyers. This platform will provide farmers with a complete digital solution that includes automated crop disease detection, real-time weather and soil insights, and a direct marketplace to sell products without middlemen. AGRI-X will empower farmers with the tools they need to make informed decisions, increase productivity, and earn fair prices for their hard work.

Business Feasibility: This solution is highly feasible given the current agricultural and technological environment in Bangladesh. Smartphone usage and mobile internet access are increasing rapidly, even in rural farming villages. Farmers are becoming more comfortable with mobile apps and digital payment systems. The government is actively promoting digital agriculture and providing support for AgriTech startups. From a business perspective, there is growing demand from urban consumers and international buyers for high-quality, fresh, and sustainably grown agricultural products. This creates excellent market conditions for AGRI-X to succeed and grow sustainably.

Software Description and Purpose: AGRI-X is a web-based and mobile-responsive smart agriculture platform that will function as Bangladesh's first integrated solution combining AI-driven crop health monitoring, environmental intelligence, and direct farmer-to-buyer commerce. The software will allow farmers to:

1. Register securely and create detailed farm profiles
2. Upload crop images for AI-based disease detection and receive instant treatment recommendations
3. Access real-time weather forecasts, soil data, and AI-driven farming advice
4. List of agricultural products for sale directly to local and international buyers
5. Manage orders, track inventory, and record sales through a simple dashboard
6. Accept payments securely via local methods like bKash, Nagad, and Rocket
7. Receive expert guidance on sustainable and modern farming techniques

For buyers, AGRI-X will provide:

1. Easy browsing and filtering of fresh agricultural products
2. Detailed crop information including AI-verified health reports

3. Transparent farmer profiles with ratings and product history
4. Secure and convenient payment options
5. Reliable order tracking and delivery updates

The platform will include features such as multilingual support (Bangla and English), offline-capable interfaces for low-internet areas, integration with local payment gateways, automated shipping solutions, and AI-powered recommendations. AGRI-X will bridge the gap between traditional farming and modern technology, ensuring that every farmer — regardless of technical knowledge or farm size — can benefit from the digital agricultural revolution.

Problem-Solution Analysis	
Key Problems Identified	AGRI-X Solution
Limited market access for farmers Farmers cannot easily connect with urban and international buyers, remaining dependent on local middlemen who offer unfairly low prices.	**Integrated Digital Marketplace** An online platform connecting farmers directly with urban consumers, wholesalers, and international buyers, eliminating intermediaries and ensuring fair pricing for agricultural products.
Lack of early crop disease detection Most farmers rely on traditional knowledge or neighboring advice, failing to identify diseases until significant crop loss has already occurred.	**AI-Based Disease Detection & Crop Analytics** Farmers can upload crop images, and the system provides early disease identification, confidence scores, symptom descriptions, and immediate treatment recommendations.
Poor access to environmental and weather data Farmers lack reliable weather forecasts, soil health information, and irrigation guidance, making farming decisions unpredictable and risky.	**Environmental & Weather Insights** Real-time weather forecasts, soil data analysis, rainfall predictions, and irrigation recommendations help farmers optimize crop management and reduce climate-related risks.
Technical barriers for farmers Many farmers, especially in rural areas, have limited technical knowledge and cannot use complex digital platforms or understand English-only interfaces.	**User-Friendly Interface with Mobile Support** Simple, intuitive interfaces designed for users with limited technical knowledge, supporting both Bangla and English languages, and optimized for low-internet conditions.
Inefficient business management tools Farmers lack proper systems for tracking product inventory, managing orders, recording sales, and communicating with buyers efficiently.	**Complete Farm Management System** Comprehensive dashboard featuring product inventory tracking, order management, customer communication tools, sales analytics, and business performance metrics designed specifically for small-scale farmers.

Payment processing challenges Farmers have limited access to secure, reliable payment systems that work with local mobile wallets and accommodate small transaction amounts.	**Local & International Payment Integration** Secure payment processing supporting familiar local methods (bKash, Nagad, Rocket) for domestic transactions, ensuring farmers can receive payments quickly and securely without bank account requirements.
Knowledge gap in modern agriculture Farmers lack access to expert advice on fertilizers, pest control, crop rotation, soil health, and sustainable farming practices.	**AI Recommendations & Advisory Features** Farmers receive automated guidance on fertilizer usage, pest control methods, crop rotation planning, sustainable farming techniques, and best practices based on their specific crop and location.

<u>Key Benefits and Objectives:</u>

For Farmers

I. **Increased Income:** Greater earning opportunities by reaching customers directly without middlemen, ensuring farmers receive fair market prices for their agricultural products.

II. **Market Access:** Direct connections with urban consumers, wholesalers, exporters, and international buyers, reducing dependency on intermediaries who take most of the profit.

III. **Improved Productivity:** AI-driven crop health monitoring and early disease detection help farmers save crops, reduce losses, and increase overall yield.

IV. **Professional Development:** Access to modern farming tools and expert agricultural advice, regardless of educational background or technical expertise.

For Sustainable Agriculture

I. **Environmental Protection:** Promotion of eco-friendly and organic farming practices, reducing excessive pesticide and chemical fertilizer usage that harms soil and water resources.

II. **Knowledge Preservation:** A space for recording and sharing traditional farming techniques, indigenous crop varieties, and sustainable agricultural practices for future generations.

III. **Climate Resilience:** Real-time weather alerts and environmental insights help farmers adapt to climate change and reduce climate-related crop losses.

For Economic Development

I. **Food Security:** Healthier crops and reduced disease-related losses contribute to national food security and stable food supply chains.
II. **Economic Contribution:** Strengthening Bangladesh's agricultural sector, which employs nearly half the workforce, and supporting rural economic growth.
III. **Rural Empowerment:** Digital inclusion for small and marginal farmers in remote areas, giving them equal access to markets and information as large commercial farms.
IV. **Women's Empowerment:** Dedicated support for women farmers and agricultural entrepreneurs, who play a vital role in Bangladesh's farming communities but often lack access to resources and technology.

Analysis of Existing Solutions: Several platforms and methods currently operate in Bangladesh's agricultural space, but none provide a complete solution for farmers' needs.

Traditional local markets (Hat-Bazaars): Farmers sell through local weekly markets but receive low prices due to the presence of multiple intermediaries. There is no price transparency, and farmers have no access to urban or international buyers.

Generic e-commerce platforms (Daraz, Pickaboo, Evaly): These platforms focus on mass-market products, electronics, and branded goods. They are not designed for fresh agricultural products, lack features for crop disease detection, and are too complex for small farmers to use effectively.

Social media platforms (Facebook, WhatsApp, Instagram): Many farmers and buyers use Facebook groups and WhatsApp to buy and sell agricultural products. However, these methods lack structured product listings, secure payment systems, order management tools, dispute resolution mechanisms, and customer trust. There is no quality verification or farmer authentication.

Agricultural advisory apps (Various government and NGO apps): Some existing applications provide weather forecasts or farming tips, but they do not include disease detection or marketplace functionality. Most are available only in English, are difficult to navigate, and require regular internet connectivity. Farmers must use multiple separate apps for different needs, which is inconvenient and inefficient.

International AgriTech platforms (Farmers Business Network, CropIn): These platforms are designed for large commercial farms in developed countries. They are too expensive for Bangladeshi small farmers, require advanced technical knowledge, do not support local languages (Bangla), and do not integrate with local payment methods like bKash, Nagad, and Rocket.

Additionally, these existing solutions do not understand the specific agricultural context of Bangladesh—the types of crops grown, common local diseases, seasonal patterns, soil conditions, language preferences, and cultural farming practices.

Key Gaps Addressed:

AGRI-X will address several critical gaps in current solutions:

Gap : How AGRI-X Addresses It

I. **AI-Powered Disease Detection:** No existing local solution offers automated crop disease detection using image analysis. AGRI-X provides instant diagnosis and treatment recommendations.
II. **Integrated Platform:** Current solutions focus on only one aspect (marketplace OR weather OR advisory). AGRI-X combines all features into a single, unified platform.
III. **Local Payment Support:** Integration with familiar Bangladeshi payment methods including bKash, Nagad, and Rocket, plus cash on delivery for rural areas.
IV. **User Accessibility:** Interfaces suitable for farmers with limited technical skills, supporting both Bangla and English languages, and optimized for low-internet conditions.
V. **Farmer Verification & Trust:** Verified farmer profiles, AI-verified crop health reports, product ratings, and quality assurance systems building buyer confidence.
VI. **Real-Time Environmental Data:** Live weather forecasts, soil health insights, rainfall predictions, and climate alerts specifically tailored to Bangladesh's agricultural zones.
VII. **Affordable & Scalable:** Free basic tier for small farmers, with affordable premium features designed for Bangladeshi economic conditions, not international pricing.
VIII. **Cultural Integration:** Support for traditional farming knowledge, indigenous crop varieties, and local agricultural practices that preserve Bangladesh's farming heritage.

AGRI-X will fill these gaps by providing a platform designed specifically for the **Bangladeshi agricultural sector**, offering the tools, support, and features that farmers need to succeed in the digital agricultural marketplace while preserving and promoting the country's rich farming heritage and ensuring food security for future generations.

4. REQUIREMENT SPECIFICATION

4.1 System Features (Farmer Features)

Epic no	Epic	Feature no	Feature	Functional Requirements	Description
4.1.1	Farmer Authentication	**4.1.1.1**	Registration	1. The system shall allow farmers to register by providing full name, phone number, email address, password, and farm location. 2. The system shall verify the phone number or email via OTP before activating the account. 3. The system shall securely store farmer registration details in the database. 4. The system shall require acceptance of terms and privacy policy before account creation.	Enables farmers to create secure and verified accounts to access AGRI-X services
		4.1.1.2	Login	1. The system shall authenticate farmers using phone/email and password. 2. The system shall maintain secure login sessions. 3. The system	Provides secure and convenient access to farmer accounts

4.1.1				shall temporarily lock accounts after 5 failed login attempts.	
		4.1.1.3	Forgot Password	1. The system shall allow password reset via OTP verification through SMS or email. 2. The system shall enforce strong password rules during reset process. 3. The system shall expire reset tokens after 24 hours for security. 4. The system shall notify farmers of successful password changes.	Ensures secure password recovery process for farmers
4.1.2	Farmer Profile Management	**4.1.2.1**	Personal Profile	1. The system shall allow farmers to edit personal information including name and contact details. 2. The system shall enable profile picture upload with image size and format validation. 3. The system	Enables farmers to maintain accurate personal information and control privacy

4.1.2				shall provide privacy settings for profile visibility control. 4. The system shall allow farmers to request account deletion with data removal.	
		4.1.2.2	Farm Information Management	1. The system shall allow farmers to add farm size, crop types, soil category, and irrigation method. 2. The system shall enable editing and updating of farm details. 3. The system shall store historical farm data for analysis and personalized recommendations. 4. The system shall allow multiple farm locations for farmers with multiple fields.	Stores essential agricultural data for personalized recommendations and disease prediction
4.1.3	Crop Disease Detection	**4.1.3.1**	Image Upload	1. The system shall allow farmers to upload crop images for AI-based disease analysis. 2. The system shall validate	Enables farmers to submit crop images for AI-powered diagnosis

4.1.3				image format (JPEG, PNG) and size (max 10MB). 3. The system shall store uploaded images securely for analysis history. 4. The system shall support batch upload of multiple images.	
		4.1.3.2	Disease Identification	1. The system shall analyze crop images using AI models trained on Bangladeshi crop varieties. 2. The system shall identify possible diseases with confidence percentage. 3. The system shall display disease name, symptoms, and severity level. 4. The system shall provide affected crop area estimation.	Detects crop diseases automatically and accurately for early intervention
		4.1.3.3	Treatment Recommendation	1. The system shall recommend treatment and preventive measures based on detected disease. 2. The system shall suggest pesticide or organic solutions	Helps farmers take timely and informed action to save crops

				with dosage instructions. 3. The system shall provide safety instructions for chemical usage. 4. The system shall offer follow-up care recommendations.	
4.1.4	Marketplace & Selling	**4.1.4.1**	Product Listing	1. The system shall allow farmers to list agricultural products for sale. 2. The system shall enable entry of price, quantity, harvest date, and quality grade. 3. The system shall display product availability status (In Stock, Low Stock, Out of Stock). 4. The system shall allow farmers to add product images and descriptions.	Enables farmers to sell products directly to buyers without middlemen
4.1.4		**4.1.4.2**	Order Management	1. The system shall notify farmers of new orders in real-time. 2. The system shall allow order acceptance or rejection based on	Streamlines order handling and fulfillment for farmers

				stock availability. 3. The system shall update order status in real-time (Processing, Packed, Shipped, Delivered). 4. The system shall support bulk order processing for large buyers.	
4.1.5	Environmental Insights	**4.1.5.1**	Weather & Soil Data	1. The system shall provide real-time weather forecasts for farmer's location. 2. The system shall provide soil health insights and recommendations. 3. The system shall send rainfall predictions and temperature alerts. 4. The system shall provide humidity and wind speed data for crop planning.	Helps farmers make informed decisions about irrigation and crop management
4.1.6	Notifications	**4.1.6.1**	Alerts & Updates	1. The system shall send disease alerts when high-risk conditions are detected. 2. The system shall send weather warnings (heavy rain, drought, storm). 3. The system	Keeps farmers informed about critical events affecting their crops and business

				shall notify farmers about new orders and payment confirmations. 4. The system shall support SMS and in-app notifications for offline farmers.	

4.2 System Features (Buyer Features)

Epic no	**Epic**	**Feature no**	**Feature**	**Functional Requirements**	**Description**
4.2.1	Buyer Authentication	**4.2.1.1**	Registration	1. The system shall allow buyers to register by providing full name, email, phone number, password, and address. 2. The system shall verify the email or phone number before activating the account. 3. The system shall securely store registration details in the database.	Enables new buyers to create secure accounts with verified contact information
		4.2.1.2	Login	1. The system shall authenticate buyers using email/phone and password. 2. The system shall maintain secure login sessions. 3. The system shall	Provides secure and convenient access to buyer accounts

				temporarily lock accounts after 5 failed login attempts.	
		4.2.1.3	Forgot Password	1. The system shall allow password reset via email or SMS OTP verification. 2. The system shall enforce strong password rules during reset process. 3. The system shall expire reset tokens after 24 hours for security. 4. The system shall notify users of successful password changes.	Ensures secure password recovery process for buyers
4.2.1		**4.2.1.4**	Guest Checkout	1. The system shall allow non-registered users to complete purchases. 2. The system shall collect minimal required information for order processing. 3. The system shall offer account creation option after guest checkout. 4. The system shall maintain guest session security during checkout process.	Reduces barriers to purchase for first-time customers.

4.2.2	Product Browsing & Search	**4.2.2.1**	Product Browsing	1. The system shall display available agricultural products on homepage. 2. The system shall organize products into intuitive category navigation (Rice, Vegetables, Fruits, Spices). 3. The system shall showcase trending and seasonal products. 4. The system shall provide personalized recommendations based on browsing history.	Creates engaging product discovery experience for buyers
		4.2.2.2	Advanced Filtering	1. The system shall provide category-based filtering (Rice, Vegetables, Fruits, Spices). 2. The system shall enable price range filtering with customizable limits. 3. The system shall support location-based farmer filtering. 4. The system shall enable filtering by quality grade and organic certification.	Helps buyers narrow down product choices based on specific criteria

		4.2.2.3	Search Functionality	1. The system shall provide text-based product search with auto-suggestions. 2. The system shall maintain search history for registered users. 3. The system shall support voice search for users with limited literacy.	Enables buyers to find products efficiently using search methods
		4.2.2.4	Product Display	1. The system shall provide high-resolution image galleries with multiple angles. 2. The system shall display product images, description, price, stock, farmer details, and AI-verified health status. 3. The system shall allow buyers to view related products. 4. The system shall display buyer reviews and ratings.	Helps buyers make informed purchase decisions
4.2.3	Shopping Cart & Checkout	**4.2.3.1**	Cart Management	1. The system shall allow adding/removing products with quantity selection. 2. The system shall provide "Move to Wishlist" options.	Facilitates easy product selection and purchase preparation

				3. The system shall calculate total prices including taxes and shipping. 4. The system shall display estimated delivery date based on location.	
		4.2.3.2	Checkout Process	1. The system shall support both guest and registered user checkout. 2. The system shall enable delivery address selection from saved addresses. 3. The system shall allow special delivery instructions. 4. The system shall display comprehensive order summary before purchase.	Streamlines purchase completion with flexible options
		4.2.3.3	Payment Processing	1. The system shall integrate secure local payment methods (bKash, Nagad, Rocket). 2. The system shall support cash on delivery for rural areas. 3. The system shall generate payment receipts and confirmations. 4. The system shall support mobile banking and bank transfer	Provides secure and convenient payment options for all buyers

				options.seasonal product scheduling	
4.2.4	Order Management	**4.2.4.1**	Order Tracking	1. The system shall provide real-time order status updates. 2. The system shall integrate shipment tracking with courier services. 3. The system shall display estimated delivery times based on location. 4. The system shall send order confirmation, shipping, and delivery notifications.	Increases transparency and builds buyer trust
		4.2.4.2	Order History	1. The system shall maintain comprehensive order history with search functionality. 2. The system shall provide reorder functionality for previous purchases. 3. The system shall show order details including product images and farmer information.	Enables buyers to track purchase history and repeat orders easily

		4.2.4.3	Returns & Exchanges	1. The system shall process return requests with reason selection. 2. The system shall generate return labels and tracking information. 3. The system shall handle exchange processing for different products. 4. The system shall process refunds back to original payment methods.	Provides hassle-free return and exchange process for buyer satisfaction
4.2.5	Wishlist & Favorites	**4.2.5.1**	Wishlist	1. The system shall allow creation of multiple wishlists with custom names. 2. The system shall provide one-click wishlist to cart conversion. 3. The system shall send price drop alerts for wishlisted products.	Helps buyers save and organize products for future purchases
		4.2.5.2	Favorite Farmers	1. The system shall allow buyers to follow favorite farmers. 2. The system shall send notifications for new products from followed farmers. 3. The system shall show updates from	Builds buyer relationships with preferred farmers

				favorite farmers on dashboard.seasonal	
4.2.6	Reviews & Ratings	**4.2.6.1**	Product Reviews	1. The system shall provide 5-star rating system with written reviews. 2. The system shall support photo and video review uploads. 3. The system shall allow review editing within specified time limits. 4. The system shall verify that reviewers purchased the product.	Enables buyers to share experiences and help other buyers
4.2.7	Notifications	**4.2.7.1**	Order Updates	1. The system shall send order confirmation, shipping, and delivery notifications. 2. The system shall provide real-time status updates via multiple channels. 3. The system shall send payment confirmation and receipt notifications.	Keeps buyers informed about their orders through preferred channels
		4.2.7.2	Marketing Communications	1. The system shall send personalized product recommendations. 2. The system shall	Engages buyers with relevant marketing content and

				notify about sales, discounts, and special offers. 3. The system shall provide seasonal crop and harvest festival announcements. 4. The system shall support newsletter subscriptions with frequency control.	agricultural information

4.3 System Features (Admin Features)

Epic no	Epic	Feature no	Feature	Functional Requirements	Description
4.3.1	Admin Authentication	**4.3.1.1**	Access Control	1. The system shall implement role-based access control with hierarchical permissions. 2. The system shall require two-factor authentication for all admin accounts. 3. The system shall log all administrative activities for audit purposes. 4. The system shall enforce IP-based access restrictions for enhanced security.	Ensures secure and controlled access to admin functions
		4.3.1.2	Admin Management	1. The system shall enable creation of admin users with specific role assignment. 2. The system shall	Manages admin user lifecycle with proper security controls

				allow permission modification with approval workflows. 3. The system shall monitor admin activity with detailed logging. 4. The system shall support secure account deactivation procedures.	
4.3.2	User Management	**4.3.2.1**	Farmer Management	1. The system shall implement farmer registration approval workflow. 2. The system shall oversee verification process with document review. 3. The system shall monitor farmer performance through metrics. 4. The system shall manage violations with warning and correction systems	Maintains platform quality through effective farmer management
		4.3.2.2	Buyer Management	1. The system shall monitor buyer activity for suspicious behavior. 2. The system shall facilitate dispute resolution between parties. 3. The system shall implement fraud	Ensures safe and trustworthy buyer experience

				detection with automated alerts. 4. The system shall oversee customer support quality assurance.	
4.3.4	Content & Product Management	**4.3.4.1**	Quality Control	1. The system shall implement product listing approval with quality standards. 2. The system shall verify product authenticity and quality claims. 3. The system shall moderate content for appropriateness and accuracy. 4. The system shall enforce consistent quality standards across platform.	Maintains high-quality content and authentic product representation
		4.3.4.2	Platform Content	1. The system shall enable homepage customization with promotional content. 2. The system shall manage banner placement and scheduling. 3. The system shall support multilingual content management (Bangla/English). 4. The system shall curate agricultural and educational materials.and	Creates engaging platform experience with relevant agricultural content

				educational materials	
		4.3.4.3	Category Management	1. The system shall manage crop categories and subcategories. 2. The system shall enable seasonal category adjustments. 3. The system shall support adding new crop varieties and types. 4. The system shall manage featured product sections on homepage.	Organizes platform content for easy navigation
4.3.5	Transaction Management	**4.3.5.1**	Order Oversight	1. The system shall monitor transactions for fraud prevention. 2. The system shall oversee payment processing with real-time monitoring. 3. The system shall mediate disputes with fair resolution procedures. 4. The system shall ensure quality assurance in order fulfillment.	Maintains transaction integrity and resolves conflicts fairly
		4.3.5.2	Financial Operations	1. The system shall manage commission structures with flexible rates. 2. The system shall track platform revenue with	Manages platform finances efficiently and transparently

				comprehensive reporting. 3. The system shall process farmer payouts with automated scheduling. 4. The system shall generate financial statements for audit purposes.	
4.3.6	Analytics & Reporting	**4.3.6.1**	Performance Analytics	1. The system shall analyze user engagement across platform features. 2. The system shall track sales performance by product, category, and region. 3. The system shall monitor platform usage for optimization opportunities. 4. The system shall identify market trends through data analysis.	Provides comprehensive insights for platform optimization
		4.3.6.2	Report Generation	1. The system shall generate daily, weekly, and monthly sales reports. 2. The system shall generate farmer performance reports. 3. The system shall generate buyer behavior analytics. 4. The system shall support custom	Enables data-driven decision making

				report creation with filters.	
4.3.7	Disease Database Management	**4.3.7.1**	Disease Information	1. The system shall allow admins to add new crop diseases to the database. 2. The system shall enable updating treatment recommendations. 3. The system shall manage disease images for AI model training. 4. The system shall review and validate disease detection accuracy.	Maintains accurate and up-to-date disease detection system
		4.3.7.2	AI Model Management	1. The system shall manage AI model versions for disease detection. 2. The system shall track model accuracy and performance metrics. 3. The system shall enable model retraining scheduling. 4. The system shall validate new models before deployment.	Ensures continuous improvement of disease detection accuracy
4.3.8	Marketing Management	**4.3.8.1**	Campaign Management	1. The system shall coordinate platform-wide promotional campaigns. 2. The system shall manage featured farmer spotlights	Drives platform growth through effective marketing initiatives

				with rotation. 3. The system shall execute seasonal marketing aligned with harvest seasons. 4. The system shall oversee advertising campaigns with budget controls..	
		4.3.8.2	Partnership Management	1. The system shall manage farmer community programs. 2. The system shall coordinate agricultural organization partnerships. 3. The system shall manage NGO and government collaboration programs.	Builds strategic partnerships for platform expansion and enhancement
4.3.8	Review & Feedback Management	**4.3.8.1**	Handle Reviews and Reports	1. The system shall allow admins to delete reviews flagged as inappropriate. 2. The system shall allow admins to resolve reported issues from users. 3. The system shall allow admins to generate sales reports for farmers and platform-wide. 4. The system shall manage customer feedback and complaints.	Keeps platform safe and improves trust

4.4 Quality Attributes/Non-Functional Requirements

1. **Performance:** Supports 5,000 concurrent users; page load ≤ 3 seconds. Expected load: 5,000 concurrent users simultaneously without performance issues.
2. **Scalability:** Supports thousands of users, products, and orders
3. **Availability:** 99.5% uptime with auto-recovery
4. **Security:** Role-based access, encryption, MFA, audit logs
5. **Usability:** Mobile-responsive, bilingual (Bangla & English)
6. **Backup & Recovery:** Daily backups, recovery within 2 hours

4.5 System Interface

5. Web-based UI for Farmers, Buyers, and Admins.
6. Responsive design for desktop and mobile devices.
7. Interfaces include:

I. Login pages
II. Farmer dashboard
III. Disease detection page
IV. Product listing page
V. Buyer marketplace page
VI. Checkout page
VII. Admin panel
VIII. Project Requirements:

<u>Hardware Requirements</u>

- Personal computer with minimum Intel Core i5 processor or equivalent
- RAM: Minimum 8 GB, 16 GB Preferable
- 20 GB free disk space
- Stable internet connection

<u>Software Requirements</u>

- Operating System: Windows 10 / Linux / macOS
- Web Browser: Google Chrome / Mozilla Firefox
- Backend: ASP.NET Core / Node.js (as applicable)

- Database: SQL Server / MySQL
- Testing Tools: Selenium, Postman, JMeter, OWASP ZAP
- Documentation Tools: MS Word / Google Docs
- Defect Tracking Tool: Jira / Trello

Network Requirements

- Reliable internet connectivity
- Secure HTTPS communication

Human Resource Requirements

- Project Manager - 1
- Developers -(2-3)
- QA Lead -1
- QA Testers -(1-2)
- End Users(Farmers/Buyers)-As needed for acceptance testing

Environment

1. **Web browser-based system**
2. **Cloud hosting**
3. **Development Environment (Dev):**

    Purpose: Used by developers for coding, unit testing, and initial integration

    Server Type: Local machine or separate cloud instance

    Processor: Minimum 4 cores

    RAM: 16 GB

    Storage: 100 GB SSD

    Operating System: Windows 10 / Linux Ubuntu 20.04

    Database: Local SQL Server / MySQL instance

    Tools Installed: Visual Studio / VS Code, Git, Postman

    Access: Developer access only

4. **Quality Assurance Environment (QA):**

Purpose: Used for functional, integration, performance, Load, and security testing

Server Type: Separate cloud instance from Dev and Production

Processor: Minimum 4 cores

RAM: 16 GB

Storage: 100 GB SSD

Operating System: Same as Production (Windows Server / Linux)

Database: QA database mirroring Production schema with test data

Web Server: IIS / Nginx / Apache configured for HTTPS

Access: QA team and limited developer access

5. **Server Configuration:**

Server Type: Dedicated / Virtual Private Server (VPS) / Cloud Instance (AWS, Azure, or Google Cloud)

Processor: Minimum 4 cores, 2.5 GHz or higher

RAM: Minimum 16 GB

Storage: Minimum 100 GB SSD

Operating System: Windows Server 2019 / Linux Ubuntu 20.04 or higher

Database Server: SQL Server / MySQL configured for production

Web Server: IIS / Nginx / Apache configured for HTTPS

Backup & Recovery: Daily automated backup with offsite storage

8. FEATURES NOT TO BE TESTED

1. Third-party payment gateway internal logic (bKash, Nagad, Rocket)-Outside project scope; controlled by third-party providers
2. Courier service internal systems-Controlled by third-party logistics providers
3. User-owned devices and browser-specific issues beyond major browsers-Not controllable by the project
4. External email/SMS provider infrastructure-Third-party responsibility

9. TESTING APPROACH

9.1 Testing Levels

- **Unit Testing:** Developers -Test individual modules like login, disease detection, cart, and payment.

- **Integration Testing:** QA team -verifies interactions between integrated modules.
- **System Testing:** QA team -tests the fully integrated system for functional and non-functional requirements.
- **Load Testing:** QA team -Evaluates system performance under expected and peak user loads (5,000 concurrent users).
- **Security Testing:** QA team -verifies data protection, access control, and vulnerability handling.
- **Acceptance Testing:** End Users (Farmers/Buyers)-validate the system against business requirements.

9.2 Test Tools

- **Manual Testing:** Used for functional, integration, system, and acceptance testing through test case execution.
- **Postman:** Used for testing REST APIs, payment integration, and backend services.
- **Selenium:** Used for automating repetitive UI test cases such as login and checkout.
- **JMeter:** Used for load and performance testing under concurrent user conditions.
- **OWASP ZAP:** Used for basic security and vulnerability testing.
- **Jira:** Used to log, track, and manage defects.

9.3 Meetings

- **Weekly QA Meetings: Weekly-** Conducted to review test progress, defects, and risks.
- **Sprint Review Meetings: End of each sprint-** Held to validate deliverables.
- **Defect Triage Meetings: As needed-** Conducted as needed to prioritize and resolve critical issues.
- **Acceptance Review Meetings: During acceptance testing-** Held with stakeholders for final approval.

10.TEST CASES / TEST ITEMS

<u>Test Case 1:</u> Farmer Registration

Project Name: AGRI-X			Test Designed by: Shohaib	
Test Case ID: TC-FARM-REG-01			Test Designed date: 05-02-2026	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Farmer Authentication			Test Execution date:	
Test Title: Verify farmer registration with valid data				
Description: Verify that a new farmer can successfully register using valid input data.				
Precondition (If any): User is not registered and is on the registration page.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Enter full name, phone, email 2. Enter password 3. Enter farm location 4. Click Register	Full Name: Rahim Miah Phone: 01125658963 Email: rahim@gmail.com Password: Rahim@123 Location: Raj Shahi	Accounts will be registered. OTP sent to phone/email Verification completed		
Post Condition: Farmer accounts are created and verified.				

<u>Test Case 2:</u> Farmer Login (Valid Credentials)

Project Name: AGRI-X			Test Designed by: Shohaib	
Test Case ID: TC-FARM-LOGIN-01			Test Designed date: 08-01-2026	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Farmer Authentication			Test Execution date:	
Test Title: Verify login with valid credentials				
Description: Verify that a farmer can successfully log in using valid phone/email and password.				
Precondition (If any): Farmer is registered and is on the login page.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Enter email/phone 2. Enter password 3. Click Login	Username: rahim@gmail.com Password: Rahim@123	Farmer will be successfully logged in. Dashboard displayed		
Post Condition: Farmer session created and login recorded.				

Test Case 3: Login Failure & Account Lock

Project Name: AGRI-X	Test Designed by: Shohaib
Test Case ID: TC-FARM-LOGIN-02	Test Designed date: 08-01-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Farmer Authentication	Test Execution date:
Test Title: Verify account lock after 5 failed attempts	
Description: Verify that the farmer account is locked after 5 consecutive failed login attempts.	
Precondition (If any): Farmer is registered and is on the login page.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Enter username 2. Enter wrong password 5 times 3. Attempt 6th login	Email: rahim@gmail.com Password: Invalid	The account will be locked. After 5th attempt, account locked. Access denied.		
Post Condition: Account temporarily locked for security.				

Test Case 4: Password Reset Functionality

Project Name: AGRI-X	Test Designed by: Shohaib
Test Case ID: TC-FARM-RESET-01	Test Designed date: 09-01-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Farmer Authentication	Test Execution date:
Test Title: Verify password reset using registered email/phone	
Description: Verify that a farmer can successfully reset their password using a registered email or phone number.	
Precondition (If any): Farmer account exists and is on the login page.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Click “Forgot Password” 2. Submit email 3. Enter OTP 4. Set new password	Registered email: rahim@gmail.com New valid password: Rahim@456	OTP sent to email/phone. Password will be updated		
Post Condition: Farmer can log in using the new password.				

<u>Test Case 5:</u> <u>Invalid Login Attempt</u>

Project Name: AGRI-X	Test Designed by: Shohaib
Test Case ID: TC-FARM-LOGIN-03	Test Designed date: 09-01-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Farmer Authentication	Test Execution date:
Test Title: Verify login with invalid credentials	
Description: Verify that the system displays an appropriate error message when a farmer attempts to log in with invalid credentials.	
Precondition (If any): Farmer is registered and is on the login page.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Enter Email 2. Enter wrong password 3. Click Login	Valid email: rahim@gmail.com Invalid Password: Wrong password	Login will be denied. Error message displayed		
Post Condition: Farmer remains logged out.				

<u>Test Case 6:</u> <u>Unauthorized Admin Access</u>

Project Name: AGRI-X	Test Designed by: Shohaib
Test Case ID: TC-ADMIN-SEC-01	Test Designed date: 09-01-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Admin Authentication	Test Execution date:
Test Title: Verify unauthorized access to admin panel	

Description: Verify that unauthorized users are prevented from accessing the admin panel.	
Precondition (If any): User is not an admin	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Access admin URL 2. Enter credentials 3. View error	Non-admin user	Security message will be shown		

Post Condition: Admin panel remains secured.

Test Case 7: Crop Disease Detection – Image Upload

Project Name: AGRI-X	Test Designed by: Shohaib
Test Case ID: TC-DISEASE-01	Test Designed date: 08-01-2026
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Crop Disease Detection	Test Execution date:
Test Title: Verify crop image upload for disease detection	
Description: Verify that a farmer can successfully upload a crop image for AI-based disease analysis.	
Precondition (If any): Farmer is logged in and is on the disease detection page.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Click "Upload-Image" 2. Select crop image 3. Click "Analyze"	Rice leaf image (JPG) Valid image format File size < 5MB	Image uploaded successfully Image preview displayed Processing started		

Post Condition: Image stored and ready for AI analysis.

Test Case 8: AI Disease Identification

Project Name: AGRI-X	Test Designed by: Mehedi
Test Case ID: TC-DISEASE-02	Test Designed date: 08-01-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Crop Disease Detection	Test Execution date:

Test Title: Verify AI-based disease identification	
Description: Verify that the system correctly identifies crop disease from uploaded image.	
Precondition (If any): Farmer has uploaded a crop image.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Upload diseased crop image 2. Wait for AI analysis 3. View results	Rice blast image	Disease name displayed Confidence level shown Symptoms described		
Post Condition: Disease detection result stored in history.				

Test Case 9: Treatment Recommendation

Project Name: AGRI-X	Test Designed by: Mehedi
Test Case ID: TC-DISEASE-03	Test Designed date: 08-01-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Crop Disease Detection	Test Execution date:
Test Title: Verify checkout using bKash	
Description: Verify that the system provides appropriate treatment recommendations after disease detection.	
Precondition (If any): Disease has been identified.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. After disease detection 2. View “Treatment" section Choose payment 3. Read safety instructions	Rice blast disease	Treatment recommendations shown Pesticide/organic options Safety instructions displayed		
Post Condition: Farmer can take informed action.				

Test Case 10: Farmer Product Listing

Project Name: AGRI-X	Test Designed by: Mehedi
Test Case ID: TC-PROD-LIST-01	Test Designed date: 08-01-2026
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Marketplace (Farmer)	Test Execution date:
Test Title: Verify product listing by farmer	
Description: Verify that a farmer can successfully list agricultural products for sale.	

Precondition (If any): Farmer is logged in and is on the "Add Product" page.

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Enter product name 2. Enter price 3. Enter quantity 4. Upload product image 5. Click "Submit"	Aman Rice 55 BDT/kg 100 kg Rice image	Product will be listed Visible in marketplace Buyers can view		

Post Condition: Product available for buyers.

Test Case 11: Buyer Product Search & Filtering

Project Name: AGRI-X	Test Designed by: Mehedi
Test Case ID: TC-SEARCH-01	Test Designed date: 08-01-2026
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Product Browsing (Buyer)	Test Execution date:
Test Title: Verify product search with filters	
Description: Verify that buyers can successfully search for products using available filters.	

Precondition (If any): Buyer is logged in and is on the marketplace page.

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Enter product name 2. Apply category filter 3. Apply price range 4. Apply location filter	"Rice" Grains 40-60 BDT	Correct items will be shown Filtered results		

	Rajshahi			
Post Condition: Filtered products displayed correctly.				

Test Case 12: Admin Approval of Product

Project Name: AGRI-X			Test Designed by: Mehedi	
Test Case ID: TC-ADMIN-APP-01			Test Designed date: 08-01-2026	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Admin Content Management			Test Execution date:	
Test Title: Verify admin product approval				
Description: Verify that the admin can review and approve a newly uploaded product successfully.				
Precondition (If any): User is logged in as admin and is on the review product page.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Review product 2. Approve product	Valid admin Product ID	The product will be published		
Post Condition: Product visible to customers.				

Test Case 13: Add to Cart

Project Name: AGRI-X			Test Designed by: Mehedi	
Test Case ID: TC-CART-01			Test Designed date: 08-01-2026	
Test Priority (Low, Medium, High): Low			Test Executed by:	
Module Name: Shopping Cart			Test Execution date:	
Test Title: Verify add product to cart				
Description: Verify that a buyer can successfully add a selected product to the shopping cart.				
Precondition (If any): Buyer is logged in and is on the product page.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Select product 2. Enter quantity 3. Click "Add to Cart" 4. View cart	Aman Rice Qty: 5 kg	The cart will be updated Cart badge shows 1 item		
Post Condition: Product stored in cart session.				

Test Case 14: Checkout & Payment (bKash)

Project Name: AGRI-X	Test Designed by: Anika
Test Case ID: TC-CHECKOUT-01	Test Designed date: 09-01-2026
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Checkout & Payment	Test Execution date:
Test Title: Verify checkout using bKash	
Description: Verify that the system prevents checkout when a product is out of stock.	
Precondition (If any): Buyer is logged in; cart contains products and is on the cart page.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Proceed to check out 2. Select delivery address 3. Choose bKash payment 4. Confirm payment	bKash Valid transaction	Order will be placed Payment receipt generated Order confirmation shown		
Post Condition: Order confirmed and payment receipt generated.				

Test Case 15: Checkout & Payment (Nagad)

Project Name: AGRI-X	Test Designed by: Anika
Test Case ID: TC-CHECKOUT-02	Test Designed date: 09-01-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Checkout & Payment	Test Execution date:
Test Title: Verify checkout using Nagad	

Description: Verify that a buyer can successfully complete the checkout process using Nagad payment.	
Precondition (If any): Buyer is logged in, cart contains products, and is on the cart page.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Proceed to check out 2. Select delivery address 3. Choose Nagad payment	Nagad Valid transaction	Order will be placed Payment receipt generated Order confirmation shown		
Post Condition: Order confirmed and payment receipt generated.				

Test Case 16: Payment Failure Handling

Project Name: AGRI-X	Test Designed by: Anika
Test Case ID: TC-PAY-FAIL-01	Test Designed date: 09-01-2026
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Shopping Cart & Checkout	Test Execution date:
Test Title: Verify system behavior on payment failure	
Description: Verify that the system handles payment failure correctly by displaying an error message and not placing the order.	
Precondition (If any): Buyer is logged in, cart contains products and is on the cart page.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Proceed to payment 2. Select bKash/Nagad 3. Fail transaction	Valid order Failed transaction	Error will be shown Order not created Cart remains intact		
Post Condition: Order is not created and cart remains intact.				

Test Case 17: Order Tracking

Project Name: AGRI-X	Test Designed by: Anika
Test Case ID: TC-ORDER-TRACK-01	Test Designed date: 09-01-2026
Test Priority (Low, Medium, High): Medium	Test Executed by:

Module Name: Order Management (Buyer)	Test Execution date:
Test Title: Verify order tracking status	
Description: Verify that a buyer can view the correct tracking status of their placed order.	
Precondition (If any): Buyer is logged in and has at least one placed order.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open order history 2. Click "Track Order" 3. View status	Order ID: ORD-001	Status will be shown Current location displayed Estimated delivery shown		
Post Condition: Order status updated correctly.				

Test Case 18: Out-of-Stock Product Restriction

Project Name: AGRI-X	Test Designed by: Anika
Test Case ID: TC-STOCK-01	Test Designed date: 09-01-2026
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Product Browsing (Buyer)	Test Execution date:
Test Title: Verify checkout restriction for out-of-stock products	
Description: Verify that the system prevents checkout when a product is out of stock.	
Precondition (If any): Buyer is logged in; product stock is zero.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. View product 2. Try to add to cart 3. Attempt checkout	Out-of-stock item Quantity = 0	"Out of Stock" label shown Add to cart button disabled Checkout blocked		
Post Condition: Order cannot be placed for out-of-stock product.				

Test Case 19: Admin Approval of Farmer Product

Project Name: AGRI-X	Test Designed by: Anika
Test Case ID: TC-ADMIN-APP-01	Test Designed date: 09-01-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Admin Content Management	Test Execution date:
Test Title: Verify admin product approval	
Description: Verify that the admin can review and approve a newly uploaded product successfully.	
Precondition (If any): Admin is logged in, product pending approval exists.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to "Pending Products" 2. Review product details 3. Click "Approve"	Product ID: P-100 Valid admin	The product will be published Status changed to "Approved" Product visible to buyers		
Post Condition: Product visible to buyers.				

Test Case 20: Load Testing (5,000 Concurrent Users)

Project Name: AGRI-X	Test Designed by: Anika
Test Case ID: TC-LOAD-01	Test Designed date: 09-01-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Performance Testing	Test Execution date:
Test Title: Verify system performance under 5,000 concurrent users	
Description: Verify that the system handles 5,000 concurrent users without performance degradation.	
Precondition (If any): System is deployed in QA environment.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Configure JMeter with 5,000 threads 2. Simulate user activities	5,000 virtual users	Response time ≤ 3 seconds No system crash		

3. Monitor response times 4. Analyze results	Login, search, add to cart Ramp-up: 60 seconds	Error rate < 1%		
Post Condition: System handles peak load successfully.				

11.ITEM PASS / FAIL CRITERIA

I. **Pass: All critical and major defects resolved, core features working as expected (farmer registration, disease detection, product listing, checkout, payment, order management).**

II. **Failure: Presence of unresolved critical defects affecting farmer login, disease detection, product listing, checkout, payment, or order management.**

12.TEST DELIVERABLES

- Test Plan Document
- Test Cases & Test Scripts
- Defect Reports
- Test Execution Reports
- Acceptance Test Report
- Performance Test Report (JMeter)
- Security Test Report (OWASP ZAP)

13. STAFFING AND TRAINING NEEDS

1. 1 QA Tester

Role	Count	Responsibility
Senior QA / QA Lead	1	Test planning, coordination, Lead-level execution (Security & high-level system testing) and reviews
QA Engineer	1	Full-stack testing, Automation testing
Total QA	2	

2. Developers assist during unit testing and bug fixing.

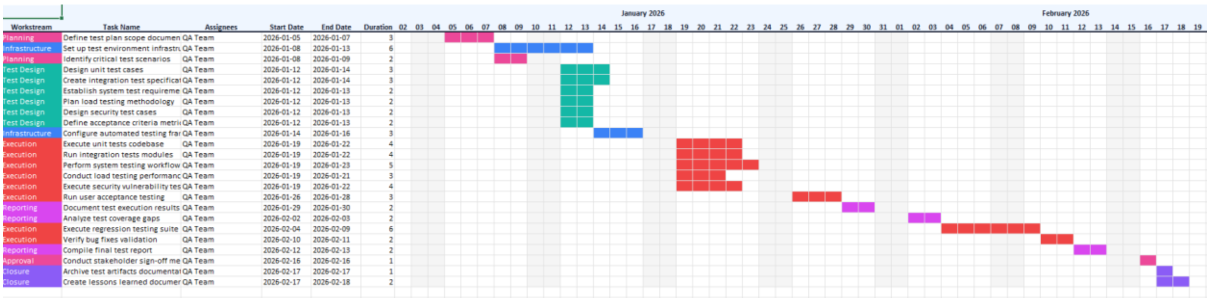
3. Training required on:

I. **AI Disease Detection Flow-**Understanding how crop image upload, AI analysis, and treatment recommendation work
II. **Payment Gateway Flow-**bKash, Nagad, Rocket integration and sandbox testing
III. **Admin Dashboard Operations-**User management, product approval, analytics, dispute handling
IV. **Load Testing (JMeter)-**Simulating 5,000 concurrent users
V. **Security Testing (OWASP ZAP)-**Vulnerability scanning and mitigation

14. RESPONSIBILITIES

Role	Responsibility
Test Manager	Overall test planning, coordination, and reporting
QA Lead	Test strategy, security testing, reviews, and team coordination
QA Tester	Test execution, defect logging, automation, load testing
Developers	Unit testing, bug fixing, integration support
Project Manager	Coordination, resource allocation, approvals
End Users (Farmers/Buyers)	Acceptance testing and feedback

15. TESTING SCHEDULE



16. PLANNING RISKS AND CONTINGENCIES

Risk ID	Risk Description	Impact	Mitigation / Contingency Plan
R1	Delay in farmer verification testing	Project timeline may be delayed	Use mock data and perform parallel testing to avoid dependency on real verification data
R2	Payment gateway issues (bKash, Nagad, Rocket)	Payment failures may affect user transactions	Use sandbox testing and multiple test scenarios before production deployment
R3	Third-party API failure or downtime (weather, SMS)	Core features dependent on API may stop working	Implement API fallback mechanisms and proper error handling; monitor API availability
R4	AI model accuracy issues	Wrong disease detection leading to crop loss	Use validated dataset, continuous retraining, and human expert verification
R5	Frequent requirement changes	Scope creep and rework may increase development time	Adopt agile methodology, maintain proper documentation, and prioritize change requests
R6	Key employee unavailability or turnover	Knowledge loss and task delays	Maintain proper documentation, code reviews, and knowledge sharing sessions
R7	Performance issues under high load (5,000 concurrent users)	Slow response time or system crash	Conduct load and stress testing using JMeter and optimize server resources

R8	Security vulnerabilities	Data breach or unauthorized access	Perform regular security testing using OWASP ZAP and enforce RBAC, encryption, and MFA
R8	Poor internet connectivity in rural areas	Farmers cannot use the platform	Implement offline support, lightweight mode, and SMS-based fallback
R9	Low farmer digital literacy	Low adoption rate	Provide training videos, Bengali interface, and field support

17.APPROVALS

Name	Designation	Signature	Date

Revision History:

Revision	Date	Updated by	Update Comments
V 1.0	11/01/2026	MD SHOHAIB ISLAM	Initial test plan created for AGRI-
V 2.0	12/01/2026	S. M MEHEDI HASAN	Testing level update (added load & security)
V 3.0	15/01/2026	S. M MEHEDI HASAN	Change formatting, added test cases
V 4.1	10/04/2026	ANIKA TAHMINA CHOWDHURY	Add server configuration, environment update
V 4.2	30-04-2026	ANIKA TAHMINA CHOWDHURY	Final version with all sections completed